APO-LORAZEPAM
Lorazepam Tablets USP 1mg
Anxiolytic-Sedative

PHARMACODYNAMICS: Apo-Lorazepam (lorazepam) is an active benzodiazepine with a depressant action on the central nervous system. It has anxiolytic and sedative properties which are of value in the symptomatic relief of pathologic anxiety in patients with anxiety disorders giving rise to significant functional disability but is not considered indicated in the management of trait anxiety.

PHARMACOKINETICS: Lorazepam is rapidly and completely absorbed after oral administration, reaching mean peak plasma levels at approximately 2 hours. Lorazepam is rapidly conjugated to a glucuronide which has no demonstrable psychopharmacological activity and is excreted mainly in the urine. The serum half life of the conjugate varied between 16 to 20 hours. Very small amounts of other metabolites and their conjugates have been isolated from urine and plasma. Most of the drug (88%) is excreted in the urine, with 75% excreted as the glucuronide. At clinically relevant concentrations, approximately 85% of lorazepam is bound to plasma proteins. Anterograde amnesia, a lack of recall of events during period of drug action, has been reported and appears to be dose-related.

INDICATIONS: Apo-Lorazepam (lorazepam) is useful for the short-term relief of manifestations of excessive anxiety in patients with anxiety neurosis. Anxiety and tension associated with the stresses of everyday life usually do not require treatment with anxiolytic drugs.

CONTRAINDICATIONS: Apo-Lorazepam (lorazepam) is contraindicated in patients with myasthenia gravis or acute narrow angle glaucoma, and in those with known hypersensitivity to benzodiazepines.

WARNINGS: Apo-Lorazepam (lorazepam) is not recommended for use in depressive neurosis or in psychotic reactions. Because of the lack of sufficient clinical experience, lorazepam is not recommended for use in patients less than 18 years of age. It should not be used in children under 6 years age. Since Apo-Lorazepam has a central nervous system depressant effect, patients should be advised against the simultaneous use of other CNS depressant drugs. Patients should also be cautioned not to take alcohol during the administration of lorazepam because of the potentiation of effects that may occur. Excessive sedation has been observed with lorazepam at standard therapeutic dosed. Therefore, patients on Apo-Lorazepam should be warned against engaging in hazardous activities requiring mental alertness and motor coordination, such as operating dangerous machinery or driving motor vehicles.
Use in Pregnancy: The safety of the use of lorazepam in pregnancy has not been established. Therefore, Apo-Lorazepam is not recommended for use during pregnancy or lactation. Several studies have suggested an increased risk of congential malformations associated with the use of the benzodiazepines, chlordiazepoxide, diazepam and meprobamate, during the first trimester of pregnancy. Since lorazepam is also a benzodiazepine derivative, its administration is rarely justified in women of child-bearing potential. If the drug is prescribed to a woman of child-bearing potential, she should be warned to contact her physician regarding discontinuation of the drug if she intends to become or suspects that she is pregnant. Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation. Apo-Lorazepam should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Anaphylaxis (severe allergic reaction) and angioedema (severe facial swelling) which can occur as early as the first time the product is taken. Complex sleep-related behaviors which may include sleep driving, making phone calls, preparing and eating food (while asleep).

PRECAUTIONS: Use in the Elderly: Elderly and debilitated patients, or those with organic brain syndrome, have been found to be prone to CNS depression after even low doses of benzodiazepines. Therefore, medication should be initiated with very low initial doses in these patients, depending on the response of the patient, in order to avoid oversedation or neurological impairment. Dependence Liability: Lorazepam should not be administered to individuals prone to drug abuse. Caution should be observed in patients who are considered to have potential for psychological dependence. It is suggested that the drug should be withdrawn gradually if it has been used in high dosage. Use in Mental and Emotional Disorders: Apo-Lorazepam (lorazepam) is not recommended for the treatment of psychotic or depressed patients. Since excitement and other paradoxical reactions can result from the use of these drugs in psychotic patients, they should not be used in ambulatory patients suspected of having psychotic tendencies. As with other anxiolytic-sedative drugs, lorazepam should not be used in patients with non-pathological anxiety. These drugs

are also not effective in patients with characterological and personality disorders or those with obsessive-compulsive neurosis. When using Apo-Lorazepam, it should be recognized that suicidal tendencies may be present and that protective measures may be required. Use in Patients with Impaired Renal or Hepatic Function: Since the liver is the most likely site of conjugation of lorazepam and since excretion of conjugated lorazepam (glucuronide) is a renal function, the usual precaution of carefully titrating the dose should be taken, should Apo-Lorazepam be used in patients with mild to moderate hepatic or renal disease. In patients for whom prolonged therapy with Apo-Lorazepam is indicated, periodic blood counts and liver function tests should be carried out.
Ricks from Concomitant Use with Opiods: Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Apo-Lorazepam with opiods. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate. If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use. If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response. If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response. Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Apo-Lorazepam is used with opioids. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the opioid have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misue, and warn them of the risk for overdose and death associated with the use of opioids (See Drug Interactions).

DRUG INTERACTIONS: If lorazepam is to be used together with other drugs acting on the CNS, careful consideration should be given to the pharmacology of the agents to be employed because of the possible potentiation of drug effects. The benzodiazepines, including lorazepam, produce CNS depressant effects when administered with such medications as barbiturates or alcohol.
Opioids: Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death. The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at $GABA_A$ sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists. Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions). Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

ADVERSE REACTIONS: The adverse reaction most frequently reported was drowsiness. Other reported adverse reactions are dizziness, weakness, fatigue and lethargy, disorientation, ataxia, anterograde amnesia, nausea, change in appetite, change in weight, depression, blurred vision and diplopia, psychomotor agitation, sleep disturbance, vomiting, sexual disturbance, headache, skin rashes, gastro- intestinal, ear, nose and throat, musculo-skeletal and respiratory disturbances. Release of hostility and other paradoxical effects, such as irritability and excitability, are known to occur with the use of benzodiazepines. In addition, hypotension, mental confusion, slurred speech, oversedation and abnormal liver and kidney function tests and hematocrit values have been reported with these drugs.

SYMPTOMS AND TREATMENT OF OVERDOSAGE: Symptoms: With benzodiazepines, including lorazepam, symptoms of mild overdosage include drowsiness, mental confusion and lethargy. In more serious overdoses, symptoms may include ataxia, hypotonia, hypotension hypnosis, Stages I and III coma, and, very rarely, death. Treatment: In the case of an oral overdose, if vomiting has not occurred spontaneously and the patient is fully awake, emesis may be induced with syrup of ipecac 20-30ml. Gastric lavage should be instituted as soon as possible and 50-100g of activated charcoal should be introduced to and left in the stomach. General supportive therapy should be instituted as indicated. Vital signs and fluid balance should be carefully monitored. An adequate airway should be maintained and assisted respiration used as needed. With normally functioning kidneys, forced diuresis with intravenous fluids and electrolytes may accelerate elimination of benzodiazepines from the body. In addition, osmotic diuretics such as mannitol may be effective as adjunctive measures. In more critical situations, renal dialysis and exchange blood transfusions may be indicated. Published reports indicate that intravenous infusion of 0.5 to 4mg of physostigmine at the rate of 1mg/min may reverse symptoms and signs suggestive of central anticholinergic overdose (confusion, memory disturbance, visual disturbances, hallucinations, delirium); however, hazards associated with the use of physostigmine (i.e.induction of seizures) should be weighed against its possible clinical benefit.

ROUTE OF ADMINISTRATION: Oral

DOSAGE AND ADMINISTRATION:

Adults: The dosage of Apo-Lorazepam (lorazepam) must be individualized and carefully titrated in order to avoid excessive sedation or mental and motor impairment. As with other anxiolytic sedatives, short courses of treatment should usually be the rule for the symptomatic relief of disabling anxiety in psychoneurotic patients and the initial course of treatment should not last longer than one week without reassessment of the need for a limited extension. Initially, not more than one week's supply of the drug should be provided and automatic prescription renewals should not be allowed. Subsequent prescriptions, when required, should be limited to short courses of therapy.

Generalized Anxiety Disorder: The recommended initial adult daily oral dosage is 2mg in divided doses of 0.5, 0.5 and 1.0mg, or of 1mg and 1mg. The daily dosage should be carefully increased or decreased by 0.5mg depending upon tolerance and response. The usual daily dosage is 2 to 3mg. However, the optimal dosage may range from 1 to 4mg daily in individual patients. Usually, a daily dosage of 6mg should not be exceeded. In elderly and debilitated patients, the initial daily dose should not exceed 0.5mg and should be very carefully and gradually adjusted, depending upon tolerance and response.

SUPPLIED:

Apo-Lorazepam Tablets 1mg

Each white, capsule-shaped, flat-faced with bevelled edge tablets, scored and engraved "APO 1" on one side, plain on the other contains lorazepam 1mg. Available in HDPE bottle of 100 tablets and blister pack of 500 tablets and 1000 tablets Not all pack size is marketed.

Storage Conditions:

Bottle: Store below 30°C. Protect from moisture.

Manufacturer: Apotex Inc., 150 Signet Drive, Weston, Ontario, Canada M9L 1T9.

Date of Revision of Package Insert: 5 Dec2022

Apotex Inc.
150 Signet Drive, Weston
Ontario, Canada M9L IT9
Distributor : Pharmaforte (M) Sdn Bhd